ZANE AL-HAMWY

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EDUCATION

Queen's University – School of Computing — **Kingston, Ontario**
Bachelor of Computing (Honours), Minor in Mathematics — *Class of 2028*

- **Academics:** Cumulative GPA **4.20/4.30**; Dean's Honour's List With Distinction (**Top 3%**), Dean's Award of Excellence (**$400**), QTMA x Amazon Web Services (AWS) Startup Pitch Competition: **Best Design**, Scotiabank CTS Case Competition 2025: **3rd**
- **Relevant Coursework:** Calculus II (A+), Linear Algebra II (A+), Data Structures (A+), Computer Architecture (A+)

Korea University – Study Abroad — **Seoul, South Korea**
College of Informatics — *Fall 2026*

- Selected out of 40+ applicants to go on an exchange semester to Korea University in South Korea for Fall 2026

PROFESSIONAL EXPERIENCE

Scotiabank — **Toronto, Ontario**
Software Engineer Intern — *Summer 2026*

- Architected an AI document search solution in **Python** using **Microsoft Azure OpenAI** and **PostgreSQL vector databases**, enabling natural-language querying **across 10,000+** internal documents through semantic search, winning **1st place** in an innovation pitch competition
- Developed scalable web application features **using React, TypeScript,** and **REST APIs**, integrating Salesforce data to support real-time data retrieval, validation, state management, and workflow automation for internal investment banker tools
- Collaborated with a team of **10+** product managers, designers, and engineers to create C4 architecture diagrams, documenting system integrations and managing feature delivery through **Agile** workflows, backlog refinement, and Jira

Quotograph — **Waterloo, Ontario**
Software Engineer Intern — *Summer 2025*

- Implemented a **Retrieval-Augmented Generation** (RAG) pipeline using **GPT-4**, enabling construction teams to query **2,000+** zoning laws by city and accelerate project approval timelines by **40%**
- Developed **10+ REST API** in Python using **FastAPI** to manage team invitations and permissions, integrating them with **5+** reusable **React** components and improving user workflow efficiency by **65%**
- Leveraged **Supabase's** cloud-hosted database and authorization features to implement role-based access for **4+ user types** and containerized the application with **Docker** to support **AWS** deployment, cutting environment-setup time by **90%** and eliminating environment-related inconsistencies across the team

Markham Website Designs — **Markham, Ontario**
Founder; Software Developer — *Mar. 2025 - Present*

- Founded and managed a web design business, delivering scalable and mobile-friendly websites used by **1,500+** monthly users across multiple industries, implementing **SEO** optimized frameworks to improve search rankings by **25%** on average
- Leveraged **JavaScript** and **Bootstrap** to build dynamic, mobile-responsive UI components and used **Google Analytics** to regularly optimize engagement, driving an average of **1,000+** visitor sessions per month for **3** concurrent clients
- Deployed client websites on **Netlify**, configuring **HTTPS** for secure browsing and integrating custom contact forms to enable direct user inquiries, increasing lead submissions by **200%**

PROJECTS

NLP-Powered Chatbot

- Built a landlord-tenant chat simulation platform in a group for 4 with **Flask** and **Next.js** to help surface **10+** fraud indicators
- Designed and implemented **Natural Language Processing** similarity checks using **SpaCy** and cosine similarity to track user questions and match them to relevant reference questions, achieving **95%** matching accuracy and reducing unmatched queries by **50%**
- Utilized the **Google Gemini API** to generate dynamic landlord personas, cutting model hallucinations by **15%** by grounding responses in reliable reference content and increasing realism in **100+** tested chat scenarios

Bonsai (Language Learning App)

- Developed a B2B language-learning startup in a 10-member interdisciplinary team, supporting **2** languages and presenting at **AWS**
- Built reusable, **TypeScript**-based frontend components to standardize core user flows, reducing frontend code duplication by **30%** and accelerating iteration on language learning features
- Enhanced **GPT-5** integration by implementing optimized request batching in **Go**, cutting **LLM** usage costs by **20%** and enabling scalable, high-volume processing without loss of output quality

ADDITIONAL INFORMATION

- **Languages:** Python, TypeScript, SQL, JavaScript, HTML/CSS, C, Java, Bash
- **Frameworks:** React.js, Next.js, Tailwind.js, Flask, FastAPI, Node.js
- **Tools & Platforms:** Git, Google Cloud, AWS, Supabase, Vercel, PostgreSQL, GitHub Copilot, Jira, Microsoft Azure
- **Interests:** Japanese, Pokémon (butoth cards and video games), Soccer, Electric Guitar, Mentoring Highschool Students, Board-game Cafés